INSTITUTIONAL COURSE SYLLABUS TEMPLATE

Course Code: CHM 101	Course Title: General chemistry
Course Instructor: **Dr. Samha Alayoubi**	Email: sayoubi@psu.edu.sa
Credit Hours: 4 (3theory+1 lab.)	Course Location: Lectures: **N115 (N Building)** Labs : **1-B02 building 105 (boys' campus)**
Scheduled Office Hours: 9-10 Mon, Tue, Wed	
Office Location: office: 121 building 102	
Co-Requisite: none	Prerequisite: none
On Campus or Online: on campus	

Mission Statement

The Department of Mathematics and Sciences is committed to offering high-quality education to lay a durable foundation for specialized degree programs at PSU. The department empowers students' academic and intellectual development by enhancing their logical thinking and analytical skills. The department adheres to the highest ethical standards and best professional practices committed to supporting the research in different scientific disciplines.

I. **Course Description:** This course is designed to provide students with foundational knowledge of chemistry. The course content encompasses basic concepts, including units and measurements, the atomic theory, principles of stoichiometry, properties of aqueous solutions, behaviors of gasses, thermochemistry, quantum numbers, periodic trends, chemical bonding, VSEPR theory, and the classification of organic compounds.

II. Course Learning Outcomes: On the successful completion of this course, students will be able to demonstrate the following

Skills	Course Learning Outcomes
Knowledge & Understanding	1.1 Recognize the basic concepts, theoretical fundamentals, and the related experimental applications in chemistry.
Skills	2.1 Predict the solution of the problem, apply suitable strategy to reach the solution, test the correctness of the solution, and interpret the results. 2.2 Apply the principles of stoichiometry in different types of chemical reactions including reactions of aqueous solutions. 2.3 Realize the main principles of thermochemistry and gases. 2.4 Interpret periodicity concepts, distinguish the types of chemical bonding, estimate the Lewis structure and molecular geometry of different chemical compounds. 2.5 Differentiate between the major classes of organic compounds and understand the IUPAC nomenclature of simple organic compounds. 2.6 Use different experimental techniques and work within a group in the laboratory, understand the objectives of the experiments, analyze data, and interpret results for deep understanding of the theoretical topics.

III. Tentative Weekly Course Schedule:

WEEK	UNIT/ TOPIC	CLO(s) alignment	STUDENT TASKS (assignments, readings, exams, quizzes, presentations, etc…)	Number of Contact hours
1 15 – 19 June **LAB #1**	Introduction 1.1 Chemistry an Overview 1.2 Scientific Method 1.3 Units of Measurements 1.4 Uncertainty in Measurements 1.5 Significant Figures and Calculations 1.9 Density 1.10 Classification of Matter. **Experiment 1:** Density and laboratory techniques.	CLO1.1+2.1 CLO 2.6	Mid-term Exam, Quizzes Homework Final Exam Lab. report Lab. midterm exam Lab final exam	6 3
2 22-26 June **LAB #2** **LAB #3**	2.2 Fundamental Chemical Laws 2.5 the Modern View of Atomic structure: An introduction. 2.6 Molecules and Ions 2.7 The Periodic Table 2.8 Naming Simple Compounds **Experiment 2:** Physical separation of Mixtures **Experiment 3:** Recrystallization technique	CLO1.1+2.1 CLO2.6 CLO2.6	Mid-term Exam Quizzes Homeworks Final Exam Lab. report Lab. midterm exam Lab final exam Lab. report Lab. midterm exam Lab final exam	6 3 3
3 29 June- 3 July **LAB #4** **LAB #5**	3.2 Atomic Masses 3.3 The Mole 3.4 Molar Masses 3.5 Learning to Solve Problems 3.6 Percent Composition of Compounds 3.7 Determining the Formula of Compounds 3.8 Chemical Equations 3.9 Balancing Chemical Equations 3.10 Stoichiometric Calculations 3.11 The concept of Limiting Reactant. **Experiment 4:** Determining the formula of hydrate **Experiment 5:** Limiting reactant	CLO1.1+2.1+2.2	Mid-term Exam Quizzes HomeWorks Final Exam	6
Week 3 **Quiz #1 (Ch. 1+2)**				
4 6-10 July	4.1 Water the common solvent 4.2 The nature of aqueous solutions 4.3 The composition of solutions: Molarity 4.4 Types of Chemical Reactions 4.5 Precipitation Reactions 4.6 Describing Reactions in Solutions 4.7 Stoichiometry of Precipitation Reactions	CLO1.1+2.1+2.2	Mid-term Exam Quizzes HomeWorks Final Exam	6

LAB #6 **LAB #7**	4.8 Acid Base Reactions **Experiment 6:** Chemicals in everyday life **Experiment 7:** Acid base titration			
MID-TERM EXAM (Ch.1+2+3+ half 4) **Tues. 8-July (12-13 PM) (unified)**				
5 13-17 July **LAB #8** **LAB #9**	5.1 Pressure 5.2 The gas laws of Boyle, Charles, and Avogadro 5.3 The Ideal Gas Law 5.4 Gas Stoichiometry 5.5 Dalton's Law of Partial Pressures: Gas Mixtures 6.1 The Nature of Energy 6.2 Enthalpy **Experiment 9:** Determining the molar mass of volatile liquid **Revision**	CLO1.1+2.1+2.3 CLO2.6	Quizzes HomeWorks Final Exam	6
17-July **Last Day for dropping one or more course(s) with a grade of "W" (1:00 PM)**				
6 20-24 July **LAB #10**	6.2 Calorimetry 6.3 Hess's Law 7.6 Quantum Numbers 7.7 Orbitals Shapes and Energies 7.8 Electron Spin and Pauli Principle 7.11 The Aufbau Principle 7.12 Periodic Trends in Atomic Properties. **Experiment 10:** Calorimetry: Heat of neutralization	CLO1.1+2.1+2.3	Quizzes HomeWorks Final Exam	6
Week 6 **Quiz #2 (Ch. 4+5)**				
7 27-31 July	8.1 Types of Chemical Bonds 8.2 Electronegativity 8.3 Polarity 8.11 Drawing Lewis Structures Exceptions of Octet Rule 8.12 Resonance 8.13 Molecular geometry and VSEPR Theory	CLO1.1+2.4	Quizzes HomeWorks Final Exam	6
Week 7 **Quiz #3 (Ch. 6+7)** **Final Lab Exams (Theory + Practical)**				
8 3-5 Aug.	22.2 Classes of Organic Compounds. 22.3 Nomenclature of Organic Compounds. 22.5 Addition and Substitution Reactions	CLO1.1+2.5	HomeWorks Final Exam	4
FINAL EXAM Sat. 10/Aug/2025 at 16:00 PM (unified)				

IV. Student Assessment & Teaching Strategies:

Domain	Assessment Task	Teaching Strategy
Knowledge	Mid-term exams + Quizzes + HomeWorks + Final Exam	Discussions Figures and animations Presentation
Skills	Mid-term Exam + Quizzes + Handworks + Final Exam+ Laboratory reports + Laboratory quizzes + Laboratory theoretical final exam + Laboratory practical exam	Discussions Figures and animations Presentation Experiment demonstration

V. Course Requirements [Whatever tasks and assignments you include in your course should be aligned with the specified learning outcomes (final learning, skills, knowledge, attitudes, and values the students leave the course with) you have defined and specified earlier. These requirements should be consistent with the Course Specification on file in the particular department.]

VI. Schedule of Assessment (Specify the schedule/date of assessment & proportion of assessment, attach related rubrics for each assessment if applicable. For examples of rubrics, visit www.irubrics.com . The schedule should be consistent with the academic calendar. Any significant changes should be avoided. If significant changes are made, academic leaders should be informed in writing, and students should be given a new schedule of assessment table).

Assessment	**Assessment Task**	**Proportion of Final Assessment**
Theory (75%)		
1	Mid-term Exam	20% 8/7/2025 from 12-1pm
2	Quizzes	10% (best 2 out of 3) Quiz#1: week 3

		Quiz#2: week 6 Quiz#3: week 7
3	Participation	5% Mainly your attendance
4	Final Exam	40% 10/8/2025 4pm
LAB. (25%)		
5	Lab. Reports	10% Weekly during the semester
6	Lab. Mid-term (Theoretical)	7.5% 31/7/2025 from 12-12:30
7	Lab. final (Practical)	7.5% Week 7 in your lab section

VII. Learning Resources

A. References

Textbook:

1) The hard copy of: Chemistry 10th edition by Steven Zumdahl and Susan Zumdahl; ISBN:ISBN-13: 978-1305957404, ISBN-10: 9781305957404
   OR
2) The electronic copy (code) of: Chemistry 10th edition by Steven Zumdahl and Susan Zumdahl; ISBN: 978 1337418171.

Lab. Manual: General Chemistry Laboratory Manual by: Ihab Shawish;
ISBN: 9786030121601
Additional Ref of the Lab: https://www.youtube.com/channel/UCzrrJQA2Z78GA6VAfVxsgLQ

Additional Reference: Chemistry, 3rd Edition; By Julia Burdge; ISBN: 978 1 259 08077
required text, essential references, recommended books, electronic materials, web sites, and other learning materials. Please indicate proper bibliographic entry).

B. Facilities Required – lecture room, computing resources, laboratories.

C. Digital Tools –MOODLE and Kahoot.

D. Learning Management System – LMS using "Moodle" is required; items such as syllabus, course materials, assignments, rubrics, and announcements will be delivered through Moodle.

E. Online Courses: Google Meet – A link will be shared prior to the session.

F. Writing and Tutoring Center – Students are highly advised to use the Writing and Tutoring Center's academic services by booking tutoring/writing appointments through the **booking system:** (https://psu.mywconline.net/)

VIII. Classroom Policies

A. Academic Integrity Policy (e.g., plagiarism or dishonesty)

"Plagiarism can be defined as unintentionally or deliberately using another person's writing or ideas as though they are one's own. Plagiarism includes, but is not limited to, copying another individual's work and taking credit for it, paraphrasing information from a source without proper documentation, and mixing one's own words with those of another author without attribution. In addition, buying a paper or project, or downloading a paper from the Internet, and submitting them as your own is also plagiarism. The penalty for academic dishonesty will bring course expulsion and failure, or even suspension" (Academic Integrity and Syllabus Acknowledgement Form).

All students are expected to submit their 'own' work and not the work of others

B. Attendance Policy:

Please, adhere to the following guidelines:

1. The University attendance policy will be strictly followed. Attendance will be taken during the first 5 minutes of the start of the lecture. Any student joining later than 10 minutes will be marked absent. Late students can still attend the class.

   In this course, the absence of ***15*** hours results in a Denied Notice (DN). (Refer to the student regulation handbook for further details.)

   Please Note: IT IS THE SOLE RESPONSIBILITY OF THE STUDENT TO KEEP SATISFYING their level of attendance. Otherwise, a "DN" grade will be granted automatically during any time in the semester without any notice from the e-register system.

2. No make-up exams will be allowed except for university accepted documents. (Refer to the student regulation handbook for further details.)

   Medical reports from either governmental or private medical centers or hospitals for a two-days absence will not be accepted as a valid reason for requesting makeup exams for midterms. Furthermore, there will be NO MAKEUP EXAMS OFFERED FOR MISSED QUIZZES

- **lab. reports Submission Policy**

1. Lab reports must be submitted on the same day as the experiment. If the lab report is submitted a day after, it will be considered late, and a deduction of 2 grades will be applied. No lab reports will be accepted beyond this deadline.
2. Students who did not attend the lab session **are not required to submit a lab report and will automatically receive a grade of zero for that particular lab**.
3. Late arrival of students exceeding 30 minutes will result in them being unable to attend the lab session, and they will receive a grade of zero for that lab.

- **General polices**

1. You are requested to take off your Abya before attending the lectures and labs.
2. Switch off your mobile phone and place it away. You are not allowed to use your phone during the lectures or labs.
3. You are requested to bring your scientific calculator and a copy of the periodic table to use during lectures and labs.
4. For safety reasons:

    a. Wearing Lab coats are **MANDATORY** during the labs.
    b. Eating or drinking in the lab is strictly forbidden.
    c. Long hair must be tied back, dangling jewelry and baggy clothing must be secured.
    d. Shoes must completely cover the foot. No sandals or flip-flops are allowed on lab days.